

AXIS INTEGRATED RESEARCH PAPERS

Working Paper AXIS2026-N1 · July 2026 | Pre-Press Edition

The Ideographic Continuum: Reconceptualizing Chinese Compound Words as Dynamic Spatio-Temporal Spectrums

[Eddie SL Chan · 陳世良]

Independent Researcher in Cognitive Linguistics & NLP
Former Software Engineer (COBOL/Prolog) & AI Annotator
Educator in Classical/Modern Chinese, English, STEM & History
Hong Kong SAR
Email: [eddieslchan@gmail.com]

ABSTRACT

Traditional structural linguistics mischaracterizes Chinese coordinate (parallel) compounds by evaluating them through Western morphological frameworks. Unlike alphabetical writing systems, Chinese does not demarcate words with spaces, and no universally accepted definition of a Chinese "word" has been established (Packard, 2000). When standard lexicography forces compounds such as 朋友 (friend), 緣分 (fate), or 宇宙 (universe) into static Western lexemes, the structural mechanics and internal kinetic energy of the ideographic script are systematically erased. This paper introduces the Character Pair Continuum Framework, which repositions parallel compounds not as fused nouns but as ideographic continuums bounded by an Initial Vector (Character A) and a Terminal Anchor (Character B). Meaning is generated kinetically as the mind traverses the spectrum between the two character brackets.

We validate this framework across four typologies: (1) Moral-Evolutionary Spectrums (朋友); (2) Positional Vector Reversals (國家 vs. 家國); (3) Spatio-Temporal Dimensional Brackets (尺寸, 鄰裡, 房屋, 宇宙); and (4) Temporal-Kinetic Timelines (緣分 vs. 緣份). We further propose Vector Field Tokenization (VFT), a theoretical NLP paradigm that models compound semantics as a line integral across a directed gradient field rather than a static dense embedding. As a theoretical framework, VFT offers a principled complement to existing radical-embedding approaches (Cao et al., 2018; Zhang & Wu, 2023), with experimental validation identified as the primary direction for future work.

Keywords: Ideographic Continuum, Chinese Morphology, Radical Mechanics, Cognitive Linguistics, Vector Field Tokenization (VFT), Lexicography, Traditional Chinese Script, Machine Translation.

Recommended citation: Chan, E. S. L. (July 2026). *The Ideographic Continuum: Reconceptualizing Chinese Compound Words as Dynamic Spatio-Temporal Spectrums*. Working Paper AXIS2026-N1. [LingBuzz]. DOI: []

Section 1: Introduction & Literature Review

1.1 The Structural Mismatch in Cross-Lingual Translation

For centuries, the translation of Chinese parallel compounds into Western alphabetical languages has suffered from a fundamental reductionism. When standard lexicography encounters compounds such as Pengyou (朋友), Yuanfen (緣分), or Yuzhou (宇宙), it forces these complex linguistic vectors into singular, static English nouns like “friend,” “fate,” or “universe.” This practice assumes a false equivalence between alphabetical morphology and ideographic typography.

Alphabetical languages use spaces between words so that discrete lexical units are visually self-evident to the reader. Chinese, on the other hand, does not follow the same practice: there is no universally accepted working definition of a “word” in Chinese, and the boundaries between morphological units remain a central, unresolved problem in Chinese linguistics (Packard, 2000). Chinese coordinate compounds, by the same token, do not operate on a one-to-one label system. By forcing a dynamic, multi-dimensional character pairing into a one-dimensional Western lexeme, the structural mechanics, historical narrative, and internal kinetic energy of the Chinese script are largely erased.

1.2 The Failure of Traditional Morphological Frameworks

Within modern Chinese linguistics, compound words are traditionally analyzed using Western morphological models, championed by scholars such as Chao (1968) and Li & Thompson (1981). These frameworks typically categorize coordinate compounds under the algebraic formula: Character A + Character B = Lexeme C.

This traditional view treats the two characters as static semantic units that merge to produce a third, independent definition. For example, standard morphology dictates that the morpheme Guó (國; state) and the morpheme Jiā (家; home) simply combine to form the stable noun Guójiā (國家; country).

This paper contests this formulaic reductionism on two grounds:

- **Semantic Flattening:** It ignores the visual primitives (radicals and other phonetic-semantic components) inherent within the characters, treating ideographs as if they were arbitrary phonetic symbols rather than packed visual data packets.
- **Directional Blindness:** It cannot account for the profound psychological and systemic shifts that occur when a character sequence is reversed (e.g., the transition from the mechanical weight of 國家機器 to the emotional resonance of 家國情懷).

Traditional linguistics treats the relationship between paired characters as a static, closed equation. In reality, the space between the characters constitutes an open, dynamic vector field.

1.3 Introducing the Character Pair Continuum Framework

To rectify this structural oversight, this paper introduces a proprietary methodology: the Character Pair Continuum Framework. This model repositions parallel Chinese compounds not as static nouns, but as ideographic continuums bounded by two conceptual brackets.

Under this framework, meaning does not reside inside Character A or Character B independently, nor does it fuse into a separate entity C. Instead, meaning is generated by the kinetic cognitive movement of the human mind as it travels across the spectrum from the initial state of the first character to the terminal anchor of the second. The paired characters act as coordinates that establish a timeline, a moral trajectory, or a spatial-scale gradient.

Section 2: Theoretical Foundations & Radical Mechanics

2.1 The Cognitive Mechanics of Ideographic Processing

To understand how parallel compounds function as continuums, we must first examine how the human brain processes the visual architecture of Chinese typography. In alphabetical writing systems, reading is primarily a phonological loop; graphemes map onto phonemes, which are then assembled linearly into an abstract semantic concept. The visual form of the word “friend” bears no structural relationship to the concept of human affinity.

In contrast, Chinese characters operate through what cognitive psychology identifies as direct dual-coding—simultaneously activating visual-spatial and semantic-phonological neural pathways (Paivio, 1986). A character is not an arbitrary signifier; it is a highly organized structural matrix comprised of sub-lexical visual primitives known as radicals (部首) alongside phonetic-semantic components that convey both sound cues and meaning hints simultaneously. These primitives serve as cognitive anchors. When two ideographs are placed adjacent to one another in a coordinate compound, their internal visual components engage in a cognitive dialogue, establishing a spatial and temporal matrix across which the reader’s eye and mind must travel (Perfetti, Liu, & Tan, 2005).

2.2 The Vector Mechanics of Radical Proximity

This framework introduces the concept of Radical Proximity Vectors. In a parallel compound, we identify two primary structural roles:

- The Initial Vector (Character A): Establishes the baseline condition, the raw material, the macro-environment, or the point of origin. It dictates where the journey begins.
- The Terminal Anchor (Character B): Establishes the resolution, the structural container, the human agency, or the localized endpoint. It dictates where the trajectory lands.

The semantic value of the word is the kinetic energy generated by the cognitive transition from the Initial Vector to the Terminal Anchor. This transition follows specific, predictable rules of radical mechanics:

- Rule I (Material-to-Agency Progression): When Character A contains raw, biological, or primordial primitives and Character B introduces high-organization or human agency primitives, the continuum tracks a progressive evolution from the primitive condition to an ennobled or anchored human state.
- Rule II (Scaling and Boundary Gradients): When the radicals and phonetic-semantic components shift from macro-scale environments to micro-scale human spaces—or vice versa—the compound acts as a lens adjusting its focal length. The mind navigates a physical gradient rather than recalling a singular static noun.

2.3 The Semantic Deficit of Alphabetic Translation

Because alphabetical words function as localized “points” of meaning on a semantic map, they are structurally insufficient for capturing the full vector paths generated by Chinese radical mechanics. This is not a claim of absolute untranslatability; rather, it is a claim about systematic semantic compression: any single Western lexeme necessarily collapses a multi-directional ideographic trajectory into a static snapshot, incurring a measurable loss of cognitive and cultural information.

When an English speaker reads the word “universe,” the mind attempts to recall a static definition: all existing matter and space considered as a whole. When a Chinese reader processes 宇宙, however, the structural mechanics actively construct the concept through its brackets: 宇 (roof covering infinite space) establishes the spatial bracket; 宙 (roof covering historical lineage) establishes the temporal bracket. The alphabetical word is a destination; the Chinese compound

is the entire voyage.

Note on DeFrancis (1989): DeFrancis argues, persuasively, that all writing systems—including Chinese—are fundamentally phonetic in nature, and that the “ideographic myth” overstates the visual-semantic dimension of Chinese characters. This paper does not dispute the phonetic dimension. Chinese characters encode both sound cues (phonetic components) and meaning hints (semantic/radical components) simultaneously, and the author’s explicit acknowledgment of “phonetic-semantic components” throughout reflects this dual structure. The present argument is that the visual-semantic layer of radicals constitutes an additional, parallel channel of meaning-making that alphabetical systems structurally lack—a position complementary to, rather than contradictory with, DeFrancis’s phonetic analysis.

Section 3: Empirical Case Studies & Typology Analysis

3.1 Typology I: Moral-Evolutionary Spectrums

Case Study: 朋友 (Péngyǒu — Friend)

The Initial Vector: 朋 (Péng). The visual architecture of 朋 consists of two identical 月 radicals side-by-side. Paleographic evidence demonstrates that in this specific semantic context, 月 is a direct evolution of the ancient pictograph for flesh or meat (肉) (Xu Shen, c. 100 CE) or the script corruption of strings of cowrie shells from oracle bone symbol / pictograph. Left unanchored by higher moral purpose, this initial vector drifts naturally into toxic, self-serving alignment—corroborated by classical idiomatic configurations such as 朋黨 (factions organized solely to protect selfish interests) and 朋比為姦 (associating closely for illicit purposes).

The Terminal Anchor: 友 (Yǒu). The visual architecture of 友 introduces a fundamental transformation, utilizing two stylized hands (又) crossing over one another. This primitive denotes active, physical, and reciprocal intervention. The structural introduction of the crossing hands stabilizes the relationship into an ennobled, collaborative bond, yielding terms of deep loyalty such as 好友, 戰友, 盟友, and 亦師亦友.

Continuum Synthesis: 朋友 is not a static noun denoting an arbitrary social status. It is a dynamic moral continuum tracking the evolution of human connection from a primitive, opportunistic assembly of flesh (朋) to the active, supportive embrace of two helping hands (友)—encapsulating the mechanics of the Western proverb: “A friend in need is a friend indeed.” A friend from the pub or a party may or may not turn into a bosom friend.

3.2 Typology II: Positional Vector Reversals

Case Study: 國家 (Guójiā) vs. 家國 (Jiāguó)

Macro-to-Micro Descent (國家 — Nation-State): In the standard configuration, 國 establishes the initial vector. Its visual architecture features a rigid outer border (口) enclosing a weapon (戈) and a defended territory, structurally signaling sovereignty, militarized defense, and administrative containment. The human scale 家 follows under the shadow of the macro-boundary. The cognitive movement flows top-down, reaching its apex in the phrase 國家機器 (the State Machine).

Micro-to-Macro Ascent (家國 — Homeland/Patria): Reversing the sequence positions 家 as the initial vector, anchoring the concept in the intimate realities of family, lineage, and personal shelter. The macro-state 國 follows not as an external regulatory cage, but as a protective, logical expansion of the home outward. The cognitive movement flows bottom-up—from individual to civilization—manifesting in the phrase 家國情懷 (deep, romanticized patriotic sentiment).

3.3 Typology III: Spatio-Temporal and Dimensional Brackets

This typology identifies compounds that utilize their paired characters as mathematical or physical coordinates to map out space, architecture, and time. Four case studies illustrate the range of this mechanism:

- 尺寸 (Chǐcùn — Dimensions/Size): 尺 (approx. one foot, anchoring macro human scale) against 寸 (approx. one inch, denoting a minute micro-increment). The compound creates a sliding measurement scale from macro-standard to micro-increment.
- 鄰裡 (Línǐ — Neighborhood): 鄰 (a cluster of five immediately adjacent households) against 裡 (a broader grid of twenty-five households cultivating shared agricultural land, Zhou Li, c. 2nd Century BCE). A sociological gradient from doorstep to district.
- 房屋 (Fángwū — Housing/Buildings): 房 (private side-rooms, the sanctuary of individual rest) against 屋 (the central public hall, the overarching roof frame). The architectural anatomy of human shelter from private sanctuary to macro structural shield.
- 宇宙 (Yǔzhòu — The Universe): 宇 (infinite spatial axis per the Huainanzi, Liu An, c. 139 BCE) against 宙 (infinite temporal axis). The literal, systemic encapsulation of Spacetime—centuries before Einstein (1920) formalized the concept in Western physics.

3.4 Typology IV: Temporal-Kinetic Timelines

Case Study: 緣分 (Yuánfèn) vs. 緣份 (Yuánfèn — Fated Affinity)

Both variants share the same Initial Vector: 緣, containing the silk radical (糸), signifying the invisible binding threads of cosmic destiny.

Trajectory A (緣分 — Meeting to Separation): Paired with 分, whose architecture of a knife (刀) causing divergence (八) introduces an expiration date. The compound tracks a bittersweet arc from meeting to separation—the universe ties the invisible knot, but the knife maps its clean, allotted limits.

Trajectory B (緣份 — Meeting to Relationship): Paired with 份, which grafts the standing person radical (亻) onto the abstract value of 分. Human agency steps onto the timeline to anchor the fleeting cosmic thread, converting a passing coincidence into a physical, realized relationship.

Section 4: Discussion & Conclusion

4.1 The Semantic Deficit of Modern Machine Translation

The empirical evidence presented across these four typologies exposes a structural challenge in contemporary Large Language Models (LLMs) and Machine Translation (MT) systems. Modern translation engines operate on vector embeddings that map words as static “points” in a high-dimensional mathematical space. When an AI system encounters 朋友, 國家, or 緣分, it queries its database for a corresponding Western point value.

As demonstrated by the Character Pair Continuum Framework, this approach results in a systematic and measurable semantic deficit. Because a Western lexeme represents a single static snapshot, it is structurally insufficient for replicating the dynamic vector fields of Chinese typography. The deficit is not merely lexical; it is architectural. When machine translation flattens 國家機器 and 家國情懷 into variants of “the nation,” or treats 緣分 and 緣份 as identical homophones, it fails to capture the cognitive simulation that a native reader experiences while

traversing from the initial character to the terminal anchor.

4.2 Implications for Cognitive Linguistics and Pedagogical Design

By recognizing parallel compounds as compressed film strips rather than labels, this framework bridges the gap between cognitive psychology and structural typography. It suggests that reading Chinese compound characters is an act of dynamic mental mapping—a continuous scaling of human emotion, physical space, and historical time.

This methodology opens new pathways for: (a) LLM Architecture—restructuring tokenizers and semantic attention heads to read Chinese compound inputs as dynamic trajectories rather than static multi-character strings; and (b) Pedagogical Frameworks—revamping second-language acquisition of Mandarin by teaching non-native speakers to read the space between character brackets, rather than forcing them to memorize isolated, flat translations.

4.3 Conclusion and Future Directions

The Character Pair Continuum Framework proposes that Chinese parallel compounds defy traditional Western morphological categorization. Meaning in Chinese typography is not a fixed destination; it is an active, cognitive voyage across a semantic spectrum. Whether tracking the moral evolution from meat/money to helping hands (朋友), the directional scale shift from state control to ancestral devotion (國家 vs. 家國), the dimensional bracketing of reality (尺寸, 鄰裡, 房屋, 宇宙), or the temporal-kinetic lifespans of human fate (緣分 vs. 緣份), these words act as fluid vectors of human experience.

Future work will expand this framework into Character Combinatorics—exploring the high-energy “particle collisions” of active primitives (CCTrio mechanics) where character intersections trigger profound directional shifts, semantic arrow vectors, and radical part-of-speech mutations. This holistic continuum model shifts our understanding of the Chinese script from a passive archive of concepts to a living, kinetic technology of narrative tracking.

Section 5: Practical Applications in Natural Language Processing (NLP)

5.1 The Architectural Deficit of Dense Vector Embeddings

Contemporary NLP architectures, including state-of-the-art Transformer-based Large Language Models, process Chinese text through an inherent architectural deficit. Current tokenizers—whether operating on Byte-Pair Encoding (BPE), WordPiece, or Unigram models—treat characters or multi-character strings as discrete, bounded lexical pieces. Once tokenized, these units are projected into high-dimensional vector spaces as static coordinates (dense vector embeddings).

Within this paradigm, the compound words 國家 (Guójiā) and 家國 (Jiāguó) are assigned independent, static vector points based on distributional semantics. Because traditional attention mechanisms calculate the geometric distance between fixed coordinates, they flatten the semantic space, remaining entirely blind to the spatial, temporal, or moral trajectories encoded within the characters’ visual primitives.

5.2 Formulating Vector Field Tokenization (VFT)

To capture the continuum mechanics of Chinese compounds algorithmically, we propose a new tokenization paradigm: Vector Field Tokenization (VFT). Instead of tokenizing a parallel compound as a single dense point, the VFT encoder decomposes the compound into an active trajectory bound by an Initial Vector ($\vec{V}_A$) and a Terminal Anchor ($\vec{A}_B$).

Mathematically, the semantic value of the continuum is modeled as a line integral across a directed gradient field, where the context of the sentence adjusts the activation path between the brackets:

$$S_C = \int_0^1 \nabla \Phi((1-t)\vec{V}_A + t \vec{A}_B) \cdot \gamma(t) dt$$

Where: $\vec{V}_A$ represents the initial sub-lexical primitive vector; $\vec{A}_B$ represents the terminal anchor structural vector; $\nabla \Phi$ represents the semantic gradient field derived from the radicals' visual and phonetic-semantic components (Perfetti et al., 2005); and $t \in [0,1]$ represents the cognitive timeline or spatial scale traversed by the system. It is worth noting that, with a linear interpolation path $\gamma(t) = (1-t)\vec{V}_A + t \vec{A}_B$, the Fundamental Theorem of Calculus for line integrals reduces the formula to $\Phi(\vec{A}_B) - \Phi(\vec{V}_A)$ —that is, the difference in semantic potential between the two character anchors. This elegant simplification grounds VFT in a tractable, computable objective: learn a scalar semantic potential function Φ over the radical-and-component embedding space.

The concept of encoding sub-character information in Chinese NLP is not without precedent. Radical-enhanced embedding approaches such as *cw2vec* (Cao et al., 2018) and sub-character tokenization models such as *SubChar* (Zhang & Wu, 2023) have demonstrated that incorporating stroke- and radical-level features measurably improves performance on downstream NLP tasks. However, these approaches treat radicals as additive feature vectors appended to character embeddings, preserving the fundamental static-point-in-space paradigm. VFT proposes a qualitatively different architecture: rather than enriching a static point with sub-character features, VFT encodes the directed relationship between two character anchors as a continuous trajectory. The semantic representation is not a coordinate at all, but the path between two coordinates—a line integral rather than a position vector. This distinction constitutes the core theoretical contribution of VFT. Empirical validation against radical-embedding baselines—including comparisons on translation quality metrics (BLEU, COMET) and semantic similarity tasks between reversed compounds such as 國家/家國 and 緣分/緣份—represents the primary direction for future experimental work.

5.3 Algorithmic Execution: The Threshold Logic Gate (能夠)

The practical utility of VFT is demonstrated during the computational processing of threshold-dependent logic. Standard NLP models regularly fail to parse the implicit structural boundaries within 能夠 (*nénggòu*), flattening the phrase to a simple auxiliary verb meaning “can” or “able to.”

Under the VFT architecture, the model executes a two-stage Threshold Logic Gate: (1) Evaluation of $\vec{V}$ _Capacity (能): the model queries the entity for intrinsic, biological, or raw computing power; (2) Evaluation of $\vec{A}$ _Threshold (夠): the model scans surrounding syntax for explicit external boundaries, dimensions, or metrics such as age, height, regulatory status, or capacity quotas.

Edge-Case Proof — The Age Metric: In the sentence “The twelve-year-old child can drive a car,” a standard model translates “can” as a uniform state of ability. A VFT-engineered model processes 能夠 as a split state: 能 = 1 (the child possesses the raw physical capacity to turn a wheel), but 夠 = 0 (the child fails to meet the external legal metric for age). The model successfully flags an Eligibility Deficit (能而不够), preserving the exact structural nuance of the text.

5.4 Re-engineering the Attention Mechanism for Machine Translation

Integrating the Character Pair Continuum Framework into Machine Translation directly resolves the endemic issue of semantic flattening during cross-lingual transfer. When a VFT-enabled MT engine encounters a continuum compound, it no longer searches for a one-to-one noun replacement. Instead, it recognizes that the source text contains a dynamic vector, triggering

a contextual expansion script in the target language:

Source Compound	Traditional LLM Output	VFT Restructured Output
國家機器	The state machine	The top-down administrative machinery of the state apparatus acting upon its constituent parts.
家國情懷	Patriotism / Homeland love	The bottom-up emotional devotion scaling from the household outward to the borders of the civilization.
緣分	Fate / Serendipity	A fated encounter destined to reach its predetermined expiration and division.
緣份	Relationship	A cosmic alignment anchored and materialized into social reality by human agency.

By translating the trajectory rather than the point, AI systems can substantially reduce the alphabetical deficit. VFT transforms machine translation from a flat word-mapping exercise into a fluid, multidimensional simulation of human experiential timelines.

References

- Cao, S., Lu, W., Zhou, J., & Li, X. (2018). cw2vec: Learning Chinese word embeddings with stroke n-gram information. *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, 32(1), 6757–6764.
- Chao, Y. R. (1968). *A grammar of spoken Chinese*. University of California Press.
- DeFrancis, J. (1989). *Visible speech: The diverse oneness of writing systems*. University of Hawaii Press.
- Einstein, A. (1920). *Relativity: The special and the general theory*. Henry Holt and Company.
- Li, C. N., & Thompson, S. A. (1981). *Mandarin Chinese: A functional reference grammar*. University of California Press.
- Liu, A. [Liu An 劉安] (comp. c. 139 BCE; trans. J. S. Major et al., 2010). *The Huainanzi (淮南子)*. Columbia University Press.
- Packard, J. L. (2000). *The morphology of Chinese: A linguistic and cognitive approach*. Cambridge University Press.
- Paivio, A. (1986). *Mental representations: A dual coding approach*. Oxford University Press.
- Perfetti, C. A., Liu, Y., & Tan, L. H. (2005). The lexical constituency model: Some implications of research on Chinese for general theories of reading. *Psychological Review*, 112(1), 43–76.
- Xu, S. [許慎] (c. 100 CE). *Shuowen Jiezi (說文解字)* [Discussing writing and explaining characters]. Han Dynasty.
- Zhang, W., & Wu, Y. (2023). Sub-character tokenization for Chinese pretrained language models. *Transactions of the Association for Computational Linguistics*, 11, 469–484.
- Zhou, G. [Attrib. Duke of Zhou / 周公旦] (c. 11th–6th Century BCE; compiled/edited c. 2nd Century BCE). *Zhou Li (周禮)* [Rites of Zhou].

Pre-Press Channels: LingBuzz (Linguistics preprints) · arXiv cs.CL (NLP/Computational Linguistics) · osf.io · zenodo.org · SSRN.com (Linguistics section). AXIS2026-F1 and AXIS2026-N1 post in parallel, July 2026, to jointly timestamp the framework terminology and the Ideographic Continuum / VFT priority claim at the same instant across all five channels.

Note on Proprietary Terms: AXIS, 300CC, MRA, MRT, ABC (Animated Bridging Component), SPS (Staff Phonetic-Semantic), GPS (Grammar Positioning/Prepositioning System), CCtrio, the 6P Handshaking Protocol (Parts/POS/Phrase/Presentation/Paragraph/Passage), TASTE, DEAN, tPontP, Scribal Convergents (隸變合流字), Visual Confusables (形近異族字), Concept Continuum (CC), and Character Colliders are original terminological contributions of this research series. First public record established upon preprint submission. All rights reserved.